TEXTBOOK EXERCISES

GB6I3Z

I. Choose the correct answer.

1 A phenomenon in which an element exists in different modification in same physical state is called
(a) isomerism (b) allotropy
(c) catenation (d) crystallinity

2 Carbon forms large number of organic compounds due to
(a) Allotropy (b) Isomerism
(c) Tetravalency (d) Catenation

3 Nandhini brings his lunch every day to school in a plastic container which has resin code number 5. The container is made of
(a) Polystyrene (b) PVC
(c) Polypropylene (d) LDPE

4 Plastics made of Polycarbonate (PC) and Acrylonitrile Butadiene Styrene (ABS) are made of resin code _____
(a) 2 (b) 5 (c) 6 (d) 7

5 Graphene is one atom thick layer of carbon obtained from
(a) diamond (b) fullerene
(c) graphite (d) gas carbon

6 The legal measures to prevent plastic pollution come under the _____ Protection Act 1988.
(a) Forest (b) Wildlife
(c) Environment (d) Human rights

II. Fill in the blanks.

1. _____ named carbon.
2. Buckminster Fullerene contains _____ carbon atoms.
3. Compounds with same molecular formula and different structural formula are known as _____.
4. _____ is a suitable solven for sulphur.
5. There are _____ plastic resin codes.

III. Match the following.

1. Alkyne	-	Bucky Ball
2. Andre Geim	-	Oxidation
3. C_{60}	-	Graphene
4. Thermocol	-	Triple bond
5. Combution	-	Polystyrene

IV. Answer briefly.

1. Differentiate graphite and diamond
2. Write all possible isomers of C_4H_{10}.
3. Carbon forms only covalent compounds. Why?
4. Define Allotrophy.
5. Why are one-time use and throwaway plastics harmful?

V. Answer in detail.

1. What is catenation? How does carbon form catenated compounds?
2. What are the chemical reactions of carbon?
3. Name the three safer resin codes of plastics and describe their features.

VI. Higher Order Thinking Skills

1. Why do carbon exist mostly in combined state?
2. When a carbon fuel burns in less aerated room, it is dangerous to stay there. Why?
3. Explain how dioxins are formed? Which plastic type they are linked to and why they are harmful to humans?
4. Yugaa wants to buy a plastic water bottle. She goes to the shop and sees four different kinds of plastic bottles with resin codes 1, 3, 5 and 7. Which one should she buy? Why?

REFERENCE BOOKS

1. Modern Inorganic Chemistry by R.D Madan
2. Fundamentals of Organic Chemistry by B.S.Bahl et.al
3. Organic Chemistry by Paula Bruise, 6th Edition

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